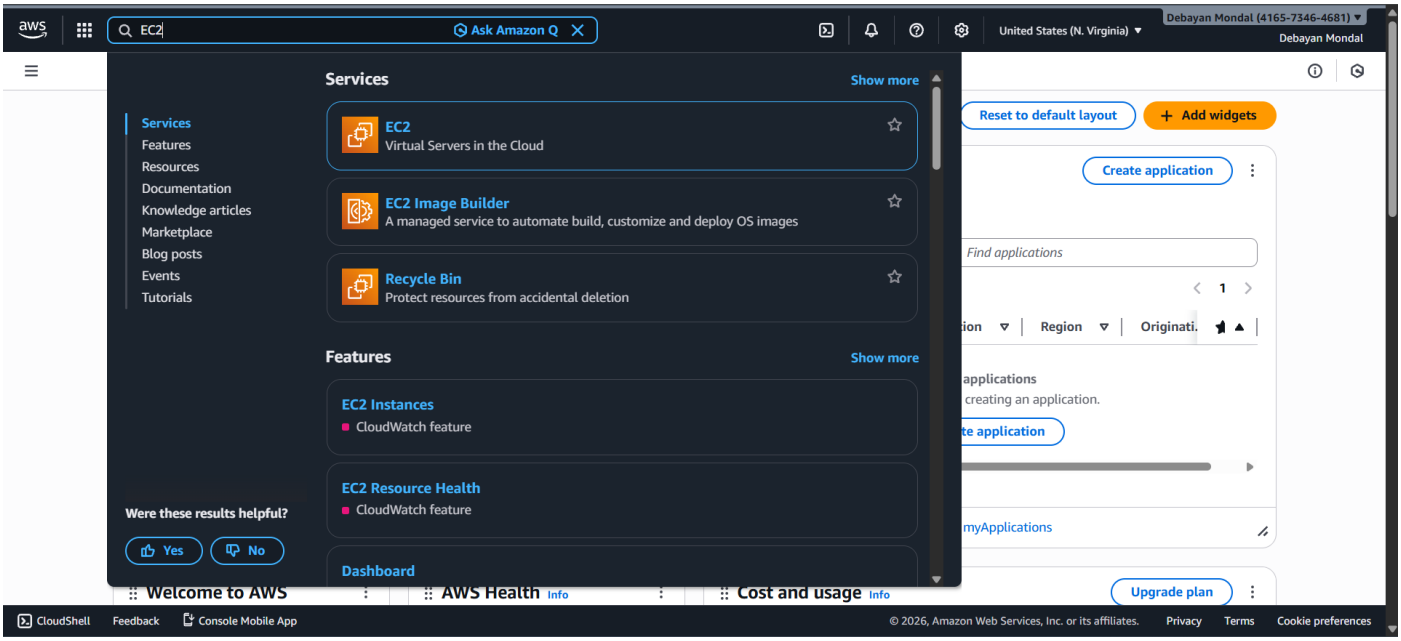
Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Assignment No.:9

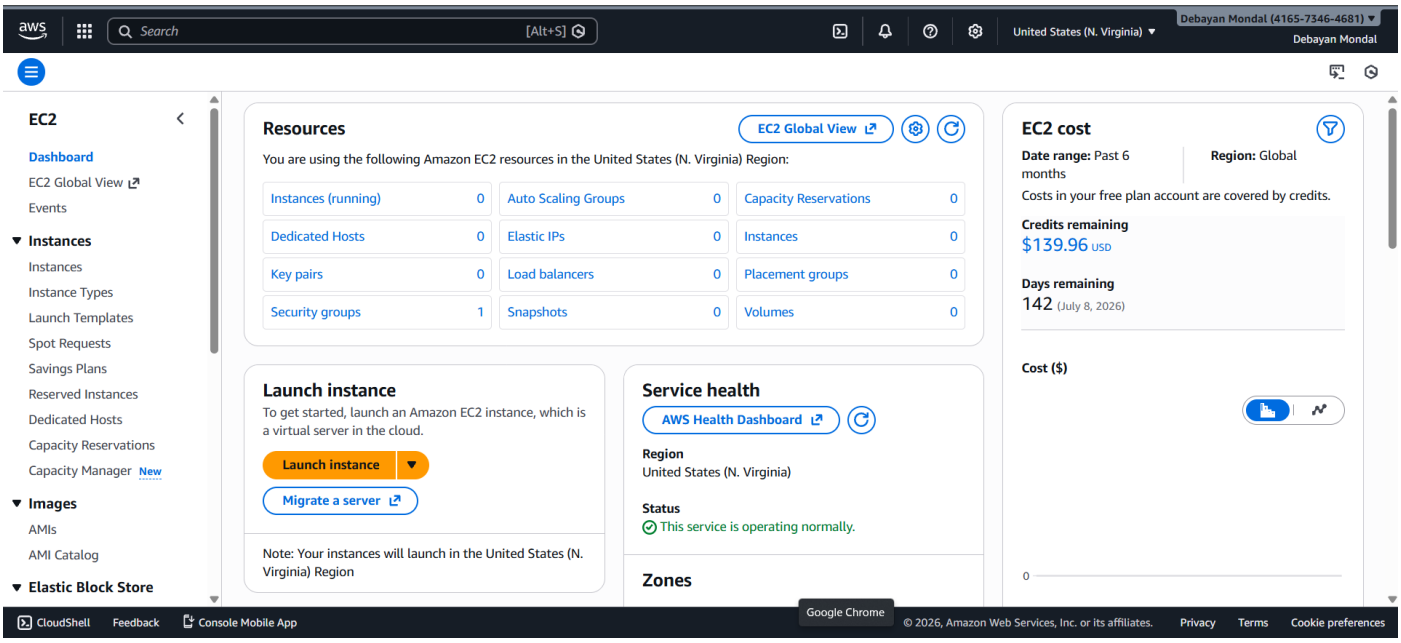
Problem Statement: Deploy a project from GitHub to EC2.

Solution:

Step 1: Search EC2 in the search bar and click on it.



Step 2: Click on **Launch Instance** button.



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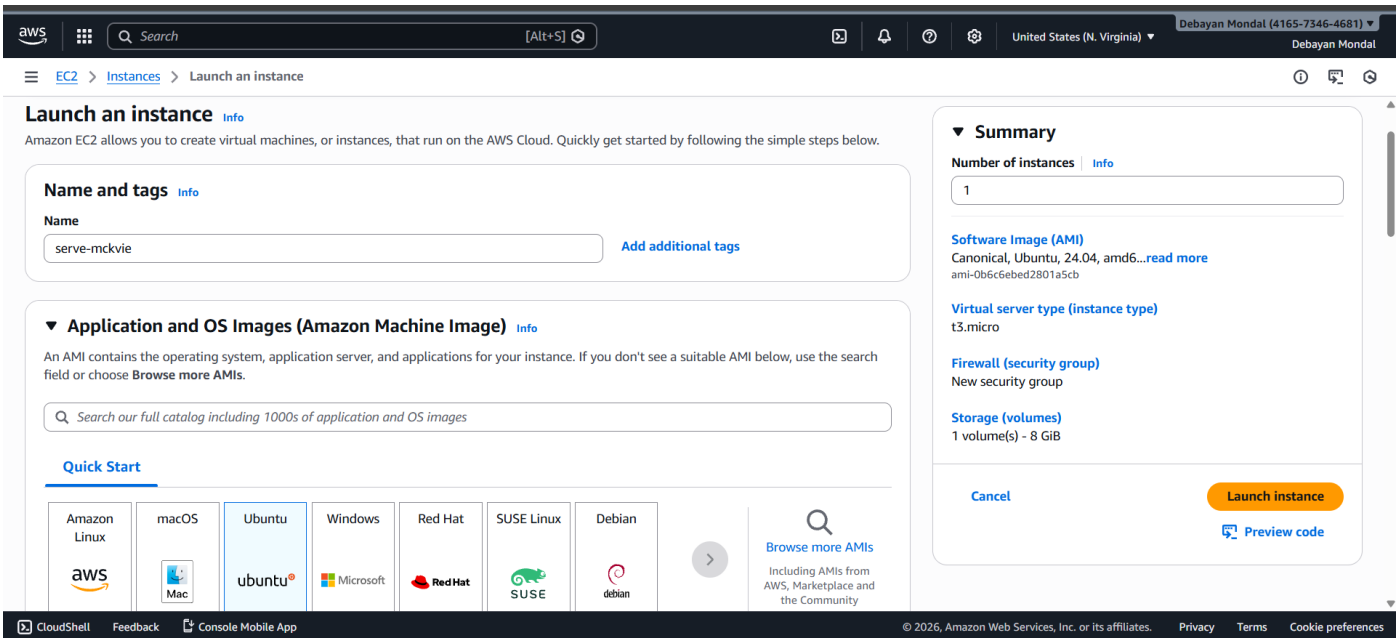
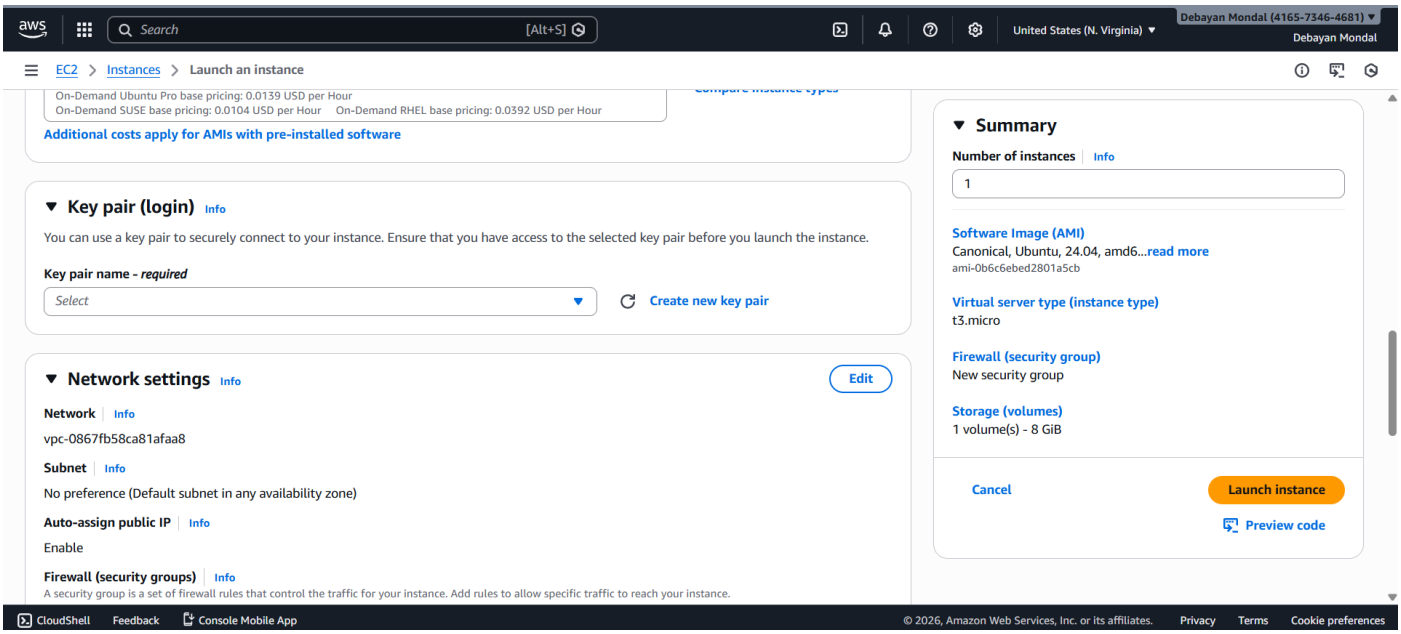
Step 3: Enter the name the Name of the instance and select the OS Ubuntu.



Step 4: Click on **Create new key pair**.



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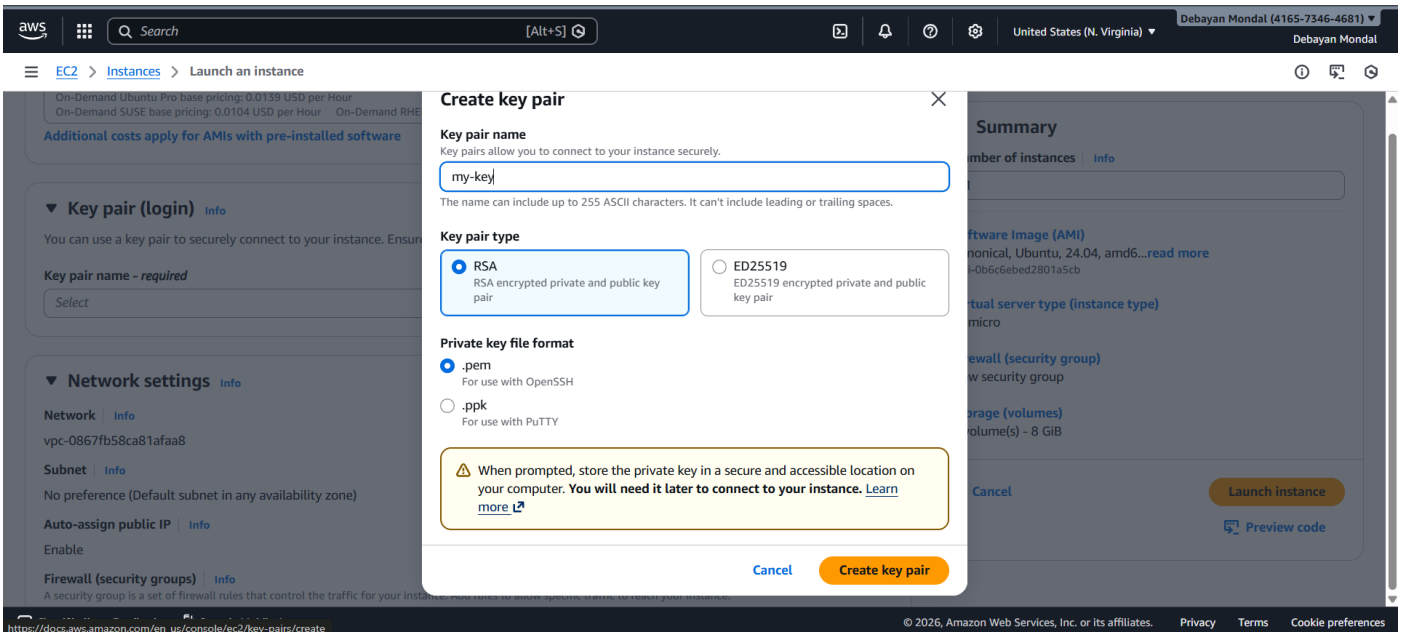
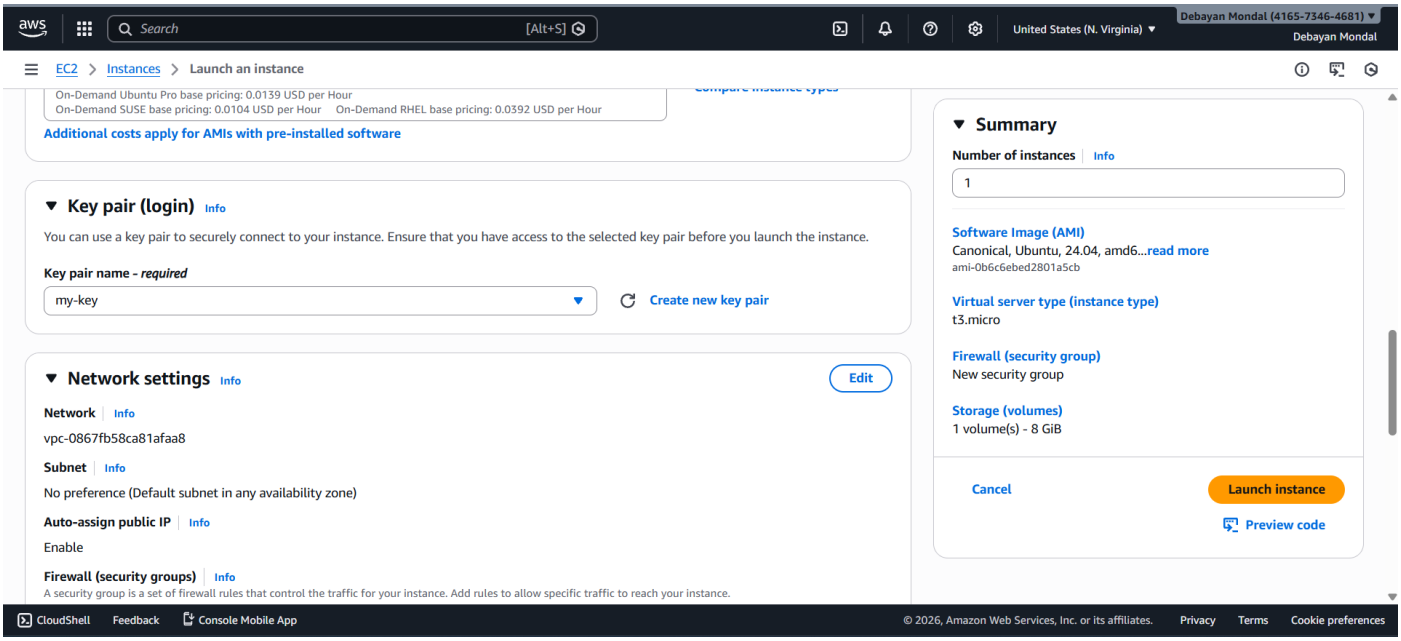
Step 5: Enter the Key pair name and click on the **Create key pair** button.



Step 6: Select the newly created key pair from the drop down.



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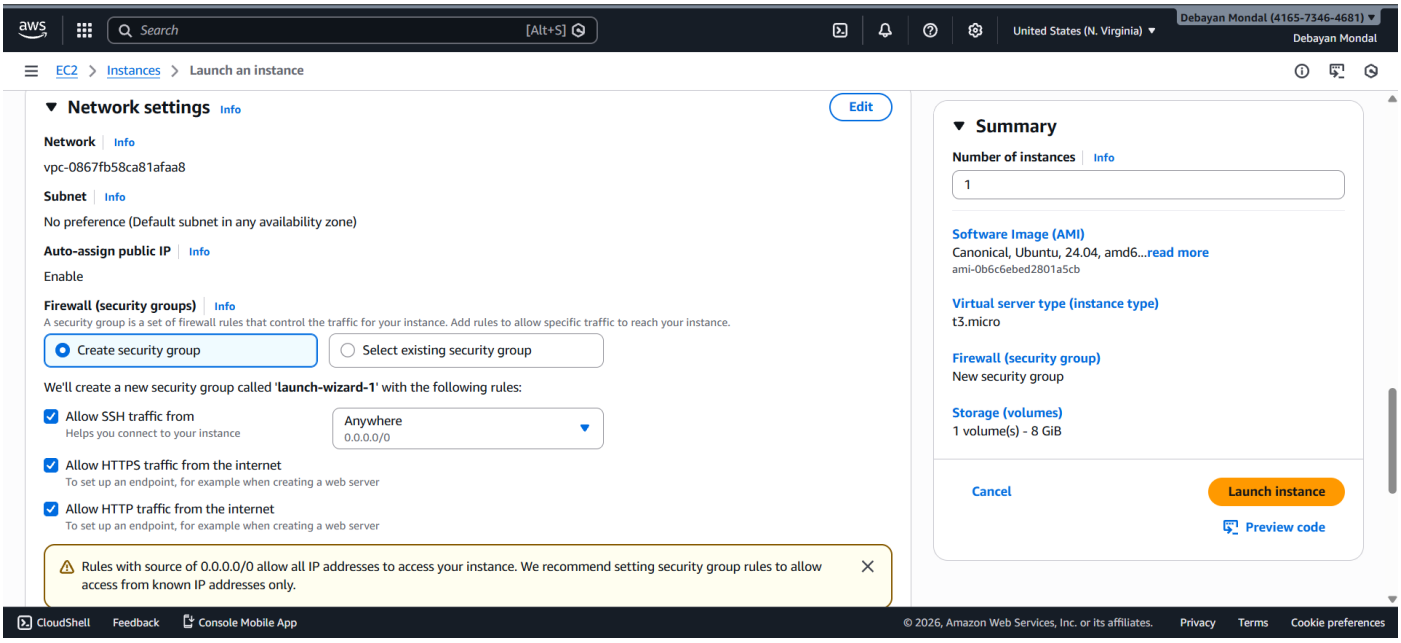
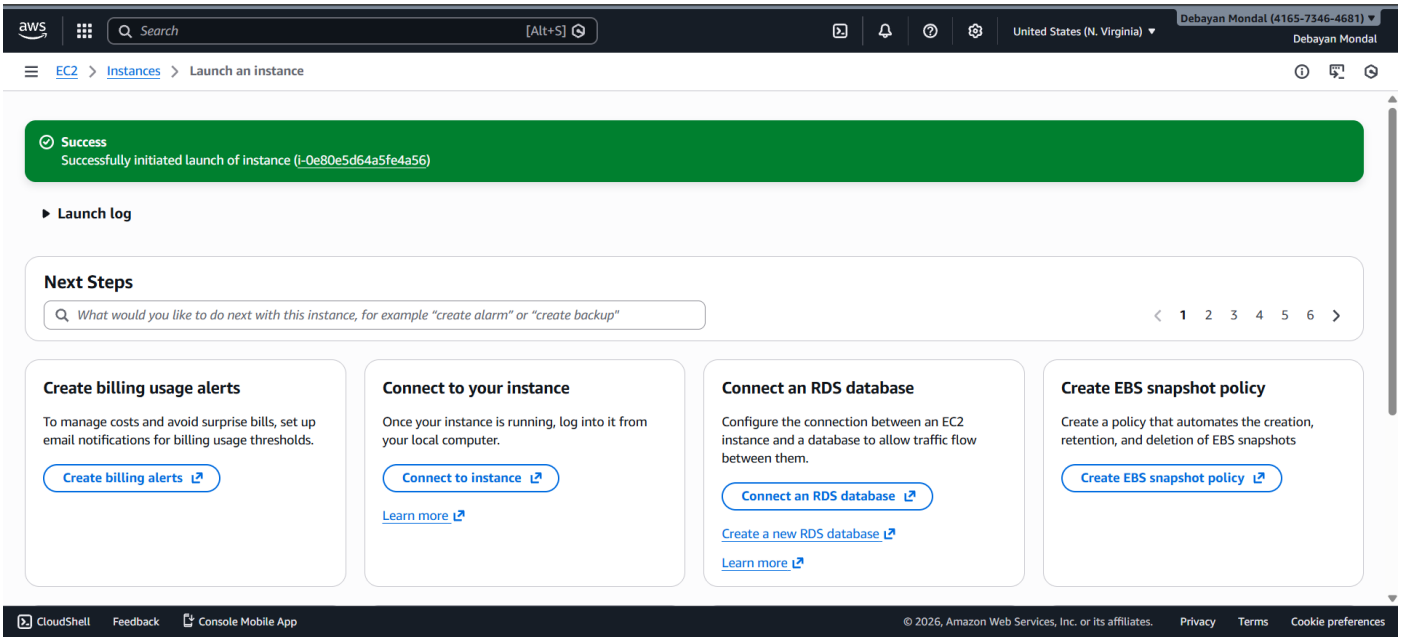
Step 7: Select **Create security group** and select the check boxes **Allow SSH traffic from, Allow HTTPS traffic from the internet, Allow HTTP traffic from the internet**. Then Select **Launch Instance** button.



Step 8: The Instance will be created successfully.



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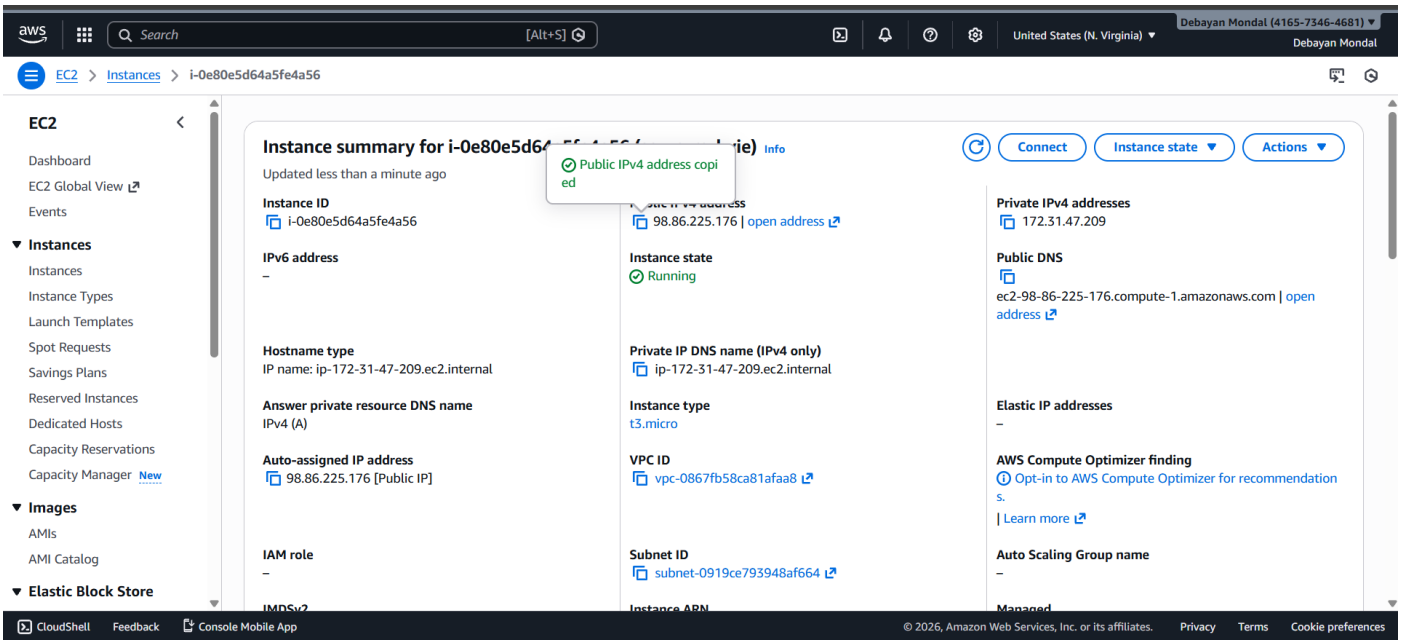
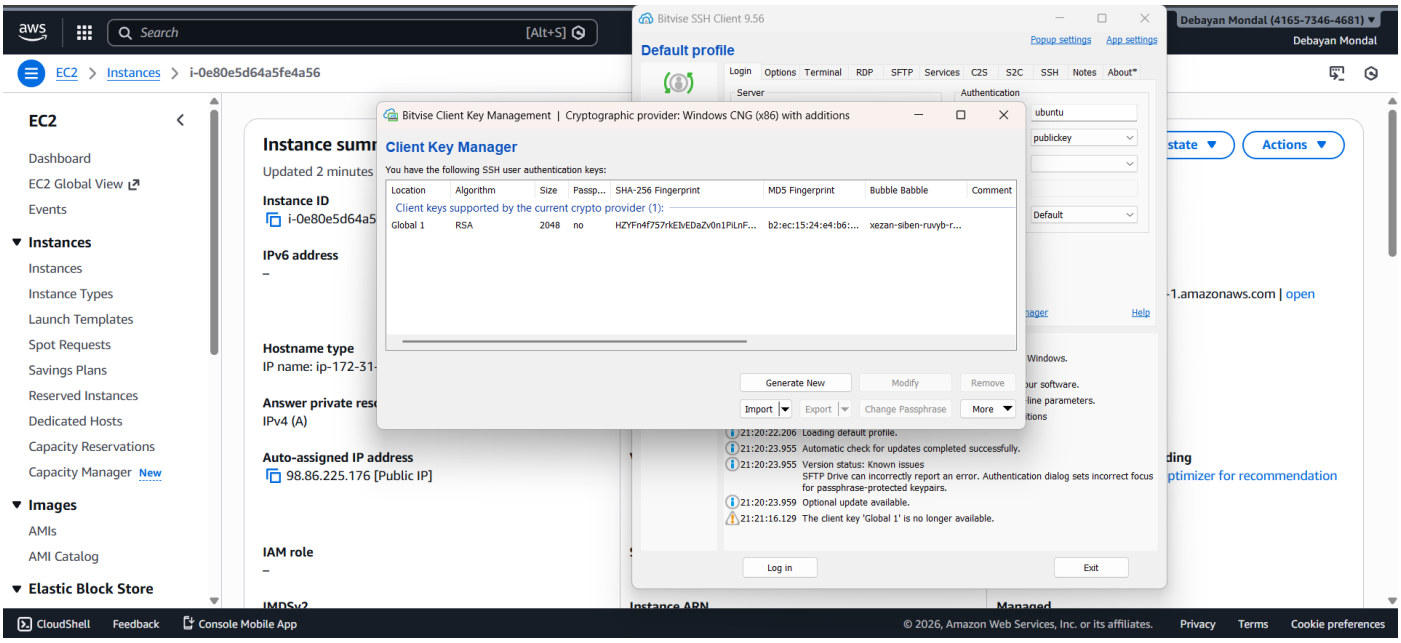
Step 9: Go to the Instance and copy the **public IPv4 address**.



Step 10: Open Bitvise and inside the Client Key Manager import the newly created key pair.



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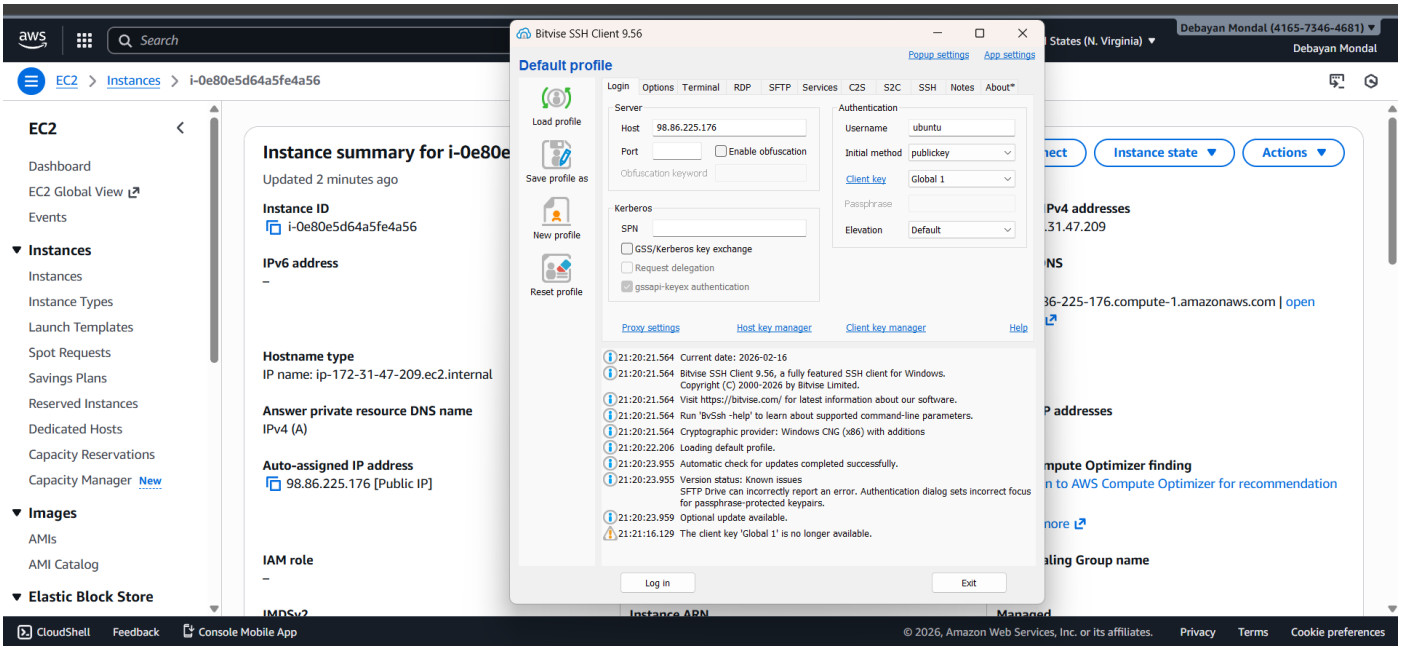
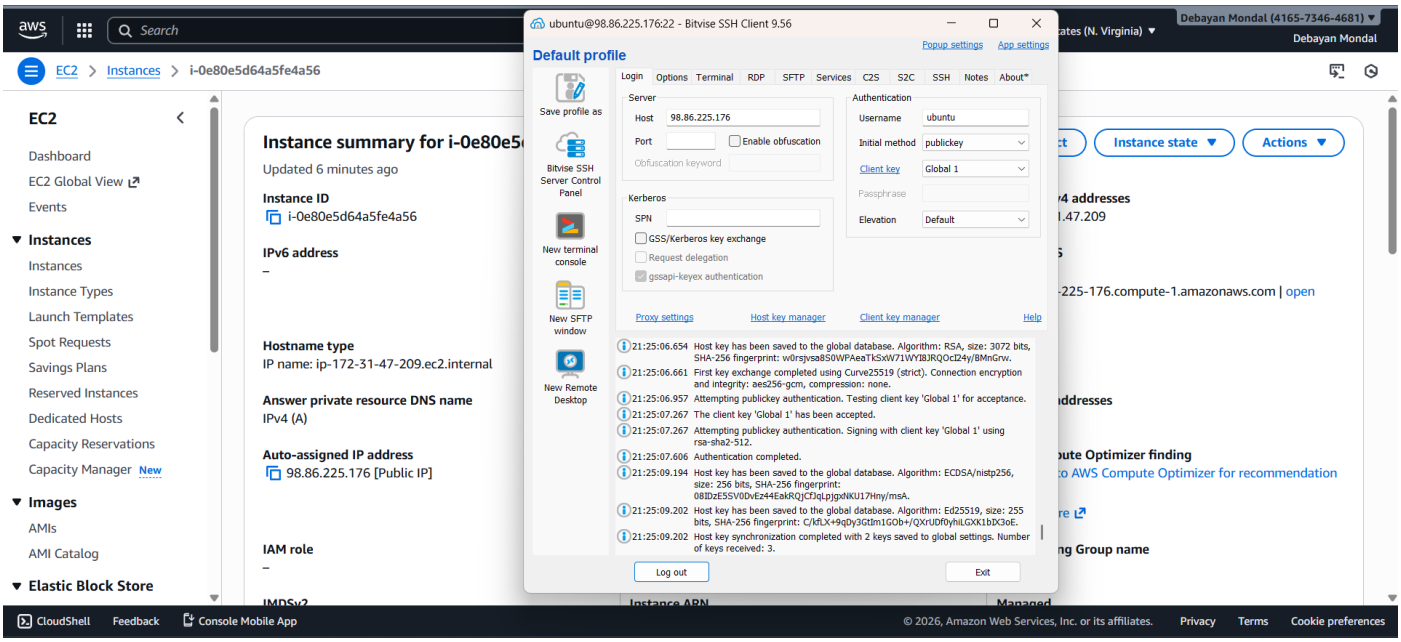
Step 11: Enter the **IPv4 adress** inside the **Host** field and Enter the **Username** field with **ubuntu**. Select the Initial method as **public key** and select the Client key as **Global 1**. Then click on the **Log in** button.



Step 12: It will be successfully connected. Click on the **New terminal console**.



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Step 13: Inside the terminal Enter the command **sudo apt-get update && sudo apt-get upgrade** to update the packages.

```
ubuntu@ip-172-31-47-209:~$ sudo apt-get update && sudo apt-get upgrade
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
```

Step 14: Enter the command **sudo apt-get install nginx** to install nginx.

```
ubuntu@ip-172-31-47-209:~$ sudo apt-get install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Step 15: We can check whether nginx is installed or not with the command **nginx -v.**

```
ubuntu@ip-172-31-47-209:~$ nginx -v
nginx version: nginx/1.24.0 (Ubuntu)
ubuntu@ip-172-31-47-209:~$
```

Step 16:Enter the command curl -sL https://deb.nodesource.com/setup_16.x|sudo -E bash -

```
ubuntu@ip-172-31-47-209:~$ curl -sL https://deb.nodesource.com/setup_16.x|sudo -E bash -

=============================================================================
=============================================================================

                           DEPRECATION WARNING

    Node.js 16.x is no longer actively supported!

  You will not receive security or critical stability updates for this version.

  You should migrate to a supported version of Node.js as soon as possible.
  Use the installation script that corresponds to the version of Node.js you
  wish to install. e.g.

   * https://deb.nodesource.com/setup_16.x — Node.js 16 "Gallium" (deprecated)
   * https://deb.nodesource.com/setup_18.x — Node.js 18 "Hydrogen" (Maintenance)
   * https://deb.nodesource.com/setup_19.x — Node.js 19 "Nineteen" (deprecated)
   * https://deb.nodesource.com/setup_20.x — Node.js 20 LTS "Iron" (recommended)
   * https://deb.nodesource.com/setup_21.x — Node.js 21 "Iron" (current)



  Please see https://github.com/nodejs/Release for details about which
  version may be appropriate for you.
```

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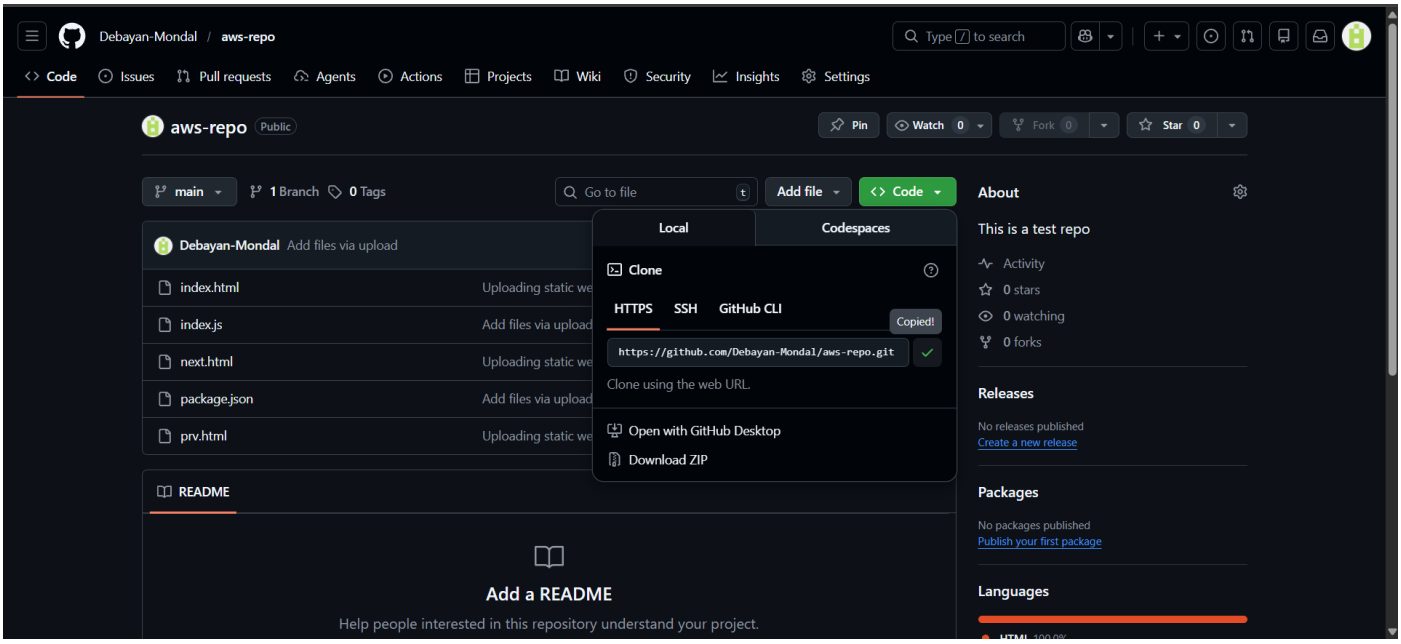
Step 17: Enter the command **sudo apt install nodejs -y**

```
ubuntu@ip-172-31-47-209:~$ sudo apt install nodejs -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  nodejs
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 27.5 MB of archives.
After this operation, 128 MB of additional disk space will be used.
Get:1 https://deb.nodesource.com/node_16.x nodistro/main amd64 nodejs amd64 16.20.2-1nodesource1 [27.5 MB]
Fetched 27.5 MB in 0s (61.3 MB/s)
Selecting previously unselected package nodejs.
(Reading database ... 71805 files and directories currently installed.)
Preparing to unpack .../nodejs_16.20.2-1nodesource1_amd64.deb ...
Unpacking nodejs (16.20.2-1nodesource1) ...
Setting up nodejs (16.20.2-1nodesource1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart getty@tty1.service
 systemctl restart networkd-dispatcher.service
 systemctl restart serial-getty@ttyS0.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 ubuntu @ session #2: sshd[1049,1162]
 ubuntu @ user manager service: systemd[1057]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-47-209:~$
```

Step 18: Go to the repo containing the project and copy its URL.



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Step 19: Enter the command **git clone <url>** to clone the website.

```
ubuntu@ip-172-31-47-209:~$ git clone https://github.com/Debayan-Mondal/aws-repo.git
Cloning into 'aws-repo'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 9 (delta 2), reused 4 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
Resolving deltas: 100% (2/2), done.
ubuntu@ip-172-31-47-209:~$
```

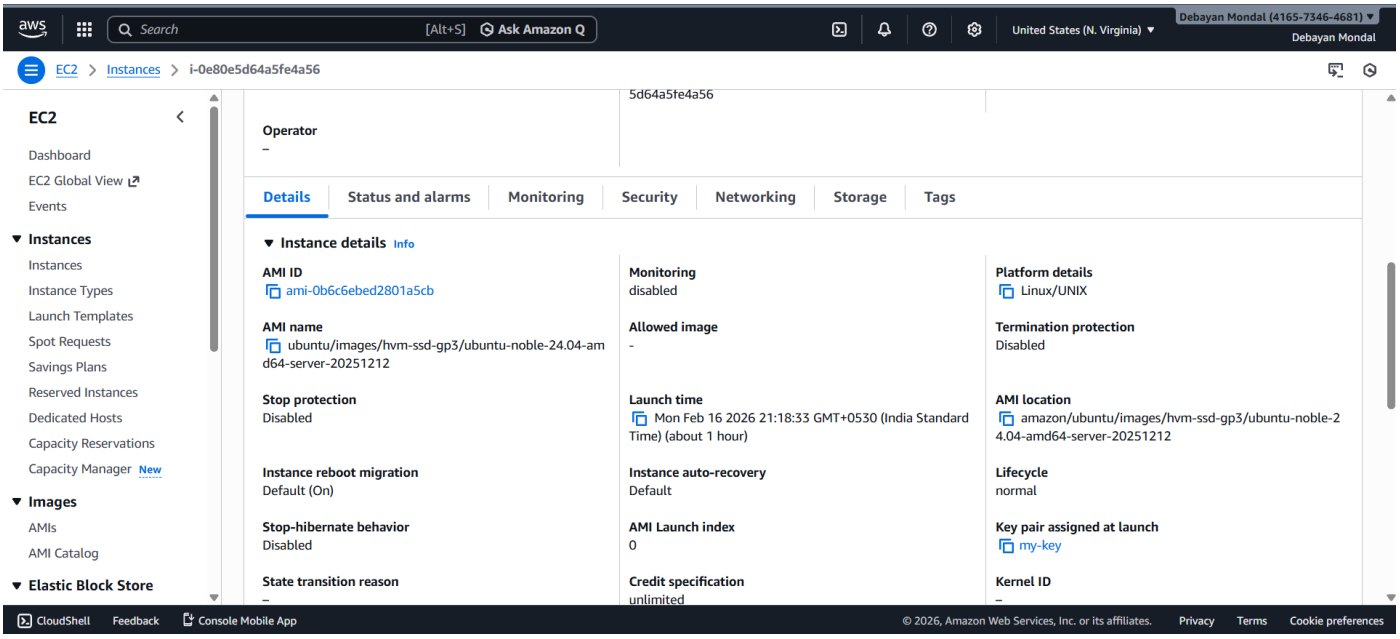
Step 20: Enter the command **npm install** to install the dependencies.

```
ubuntu@ip-172-31-47-209:~/aws-repo$ npm install

added 224 packages, and audited 225 packages in 8s

26 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 8.19.4 -> 11.10.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.10.0
npm notice Run npm install -g npm@11.10.0 to update!
npm notice
ubuntu@ip-172-31-47-209:~/aws-repo$ ls
index.html  index.js  next.html  node_modules  package-lock.json  package.json  prv.html
ubuntu@ip-172-31-47-209:~/aws-repo$
```

Step 21: Enter the command **node index.js** to start the server.

```
index.html  index.js  next.html  package.json  prv.html
ubuntu@ip-172-31-47-209:~/aws-repo/aws-repo$ node index.js
Started server
```

Step 22: Go back to the instance and Select the **Security** tab.



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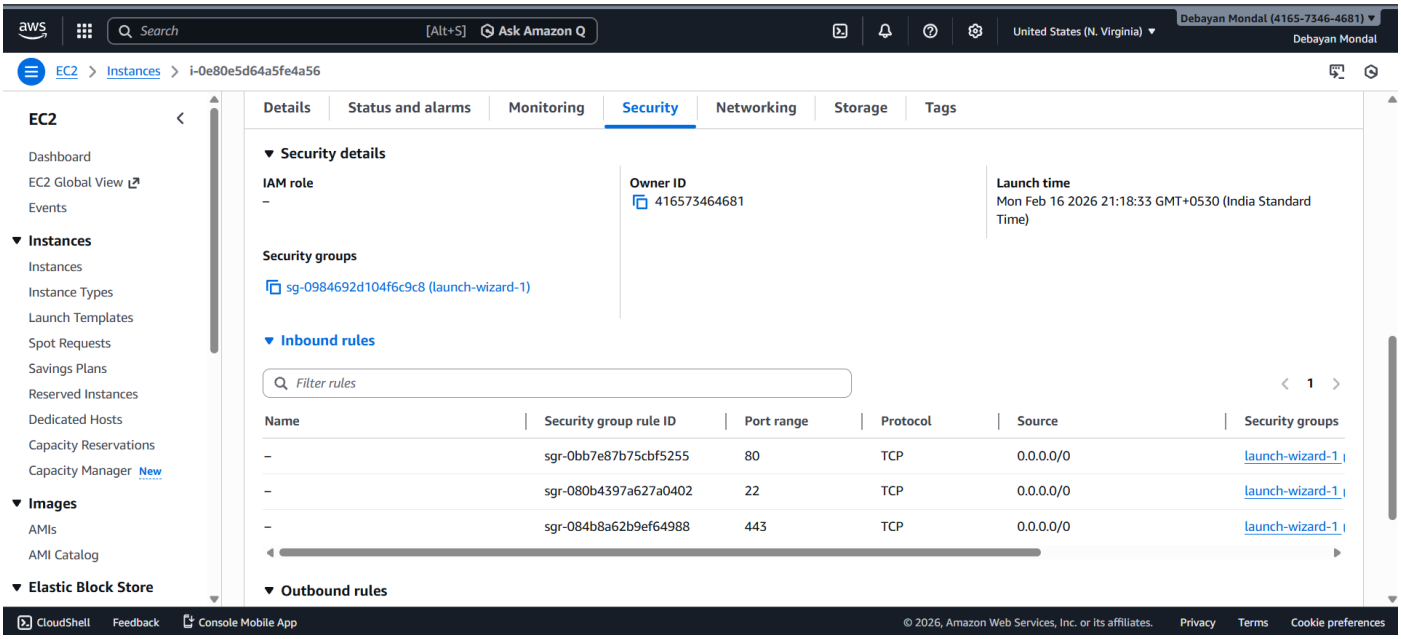
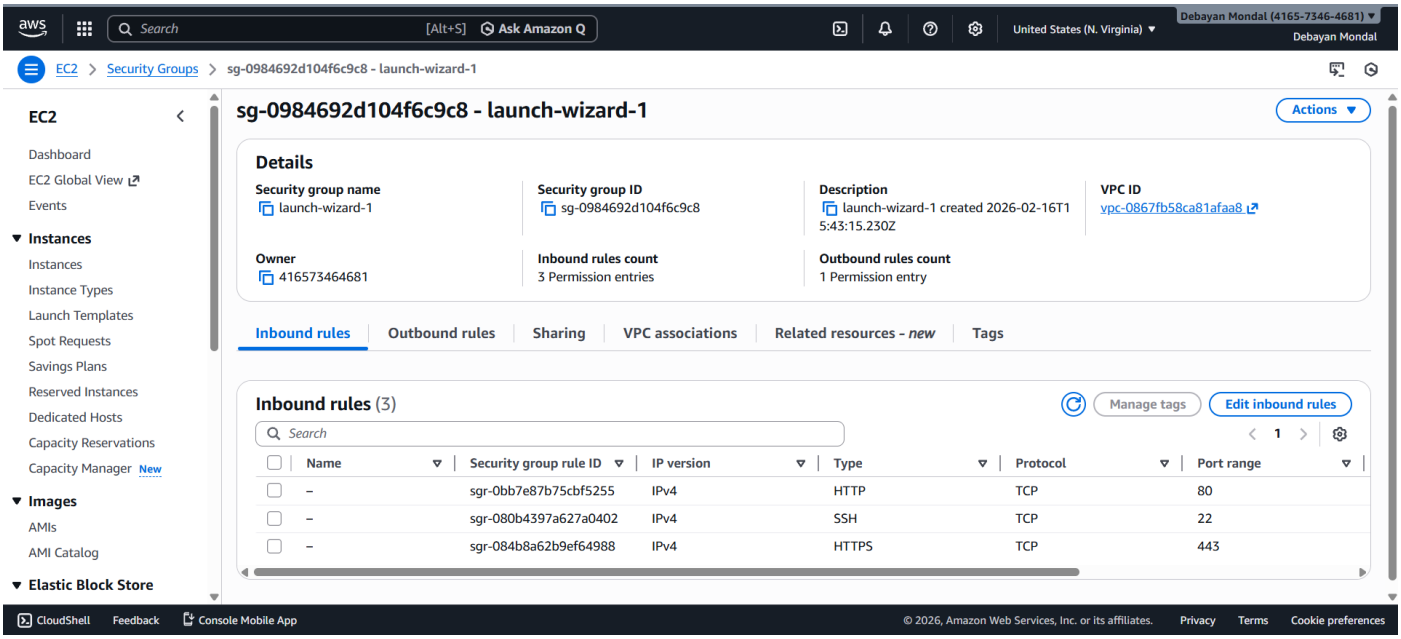
Step 23: Under the **Security** tab click on the on Security group which needs to be changed.



Step 24: Under the **Inbound rules** tab click on **Edit inbound rules** button.



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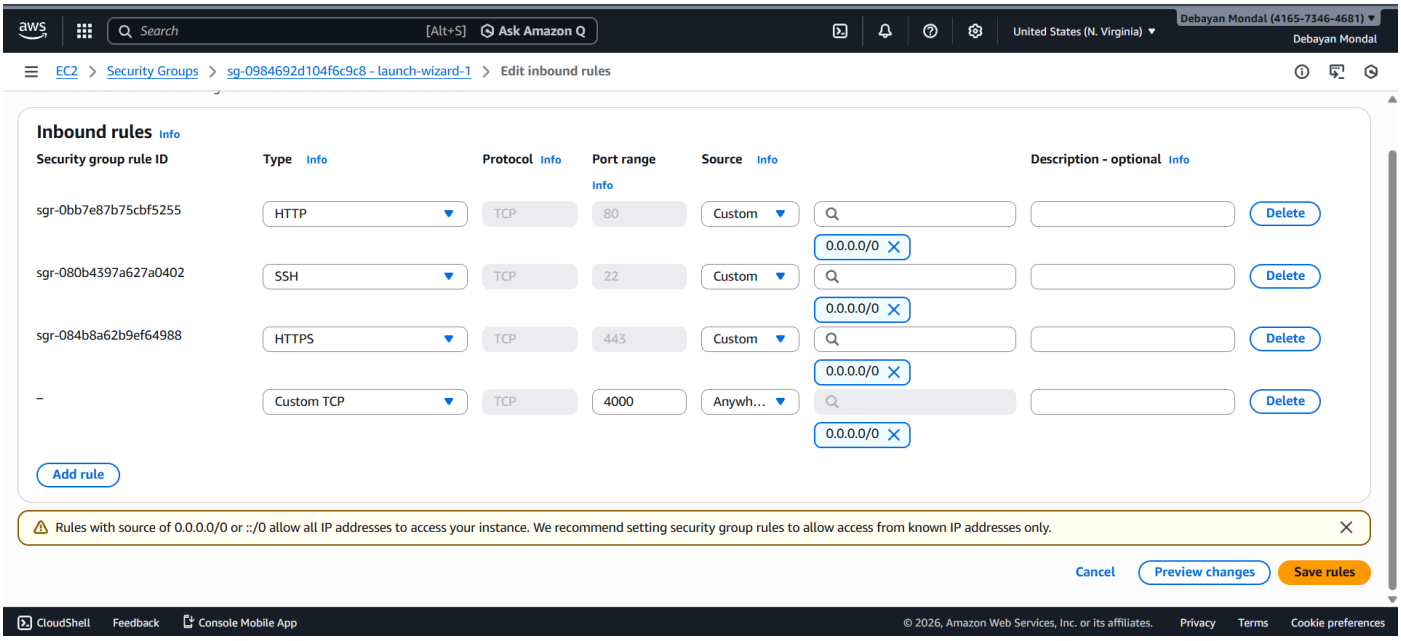
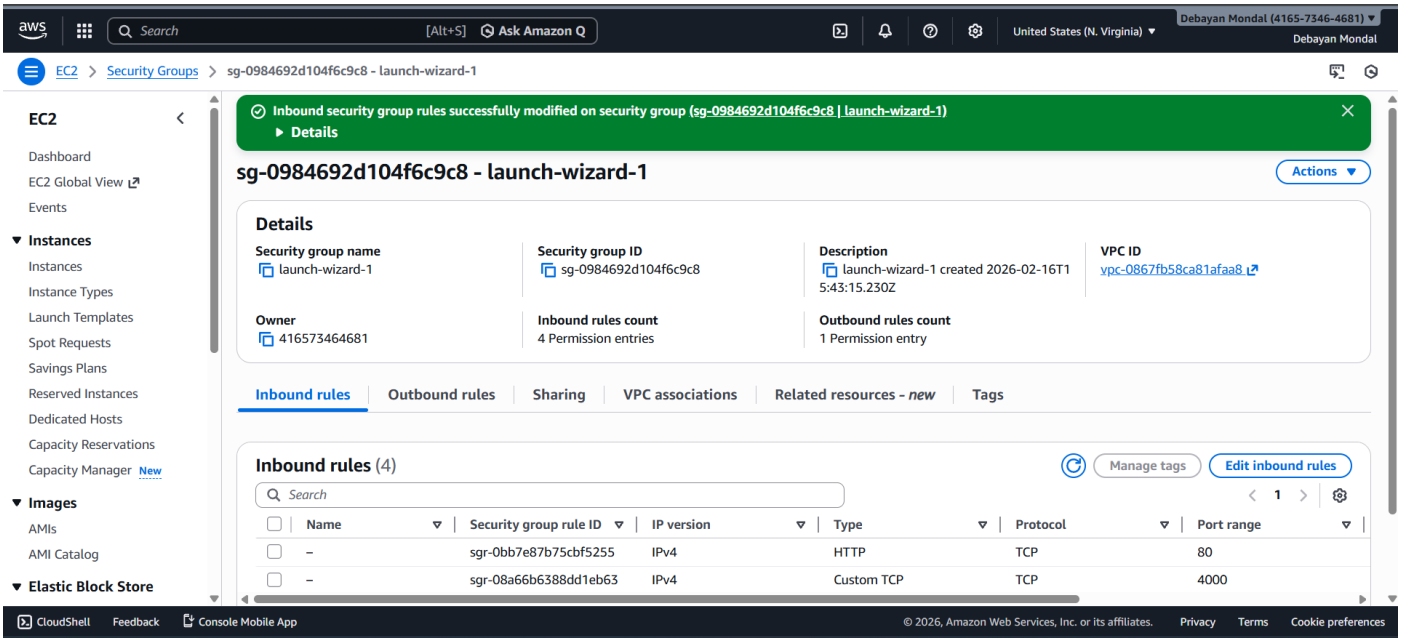
Step 25: Change the sources to **Custom** and change the types to **HTTP, SSH, HTTPS** in order and make the changes by clicking on **Save rules** button.



Step 26: The Inbound security group rules successfully modified.



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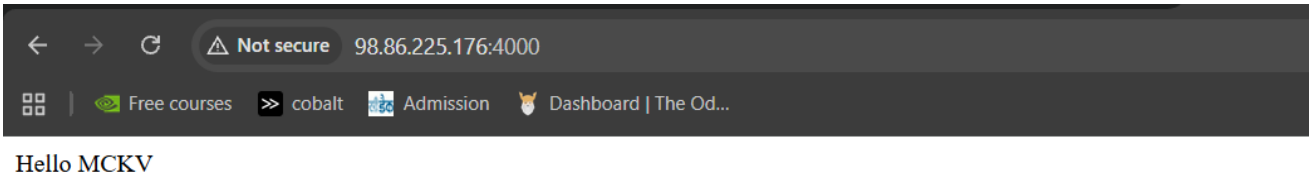
Step 27: We can now paste the public IPv4 address to see our website on a separate browser tab.